BIRDSONG

The global biogeography of passerine songs

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Although bird songs are classic models for understanding the evolution of vocal communication, their global diversity has long made the development of a unifying framework challenging. By analyzing the acoustic architecture of songs from more than 3000 passerine species worldwide, we show that this acoustic space can be structured around eight elemental motifs. The differential use of these motifs is driven by a combination of species' biological traits (social organization, morphology, and mating system) and the physics of sound propagation. In tropical rainforests, environmental filtering for transmission efficiency favors structurally simple motifs, such as flat whistles. Conversely, in temperate regions, where high population densities facilitate close-range communication and short breeding seasons intensify sexual selection, the balance shifts toward complex, information-rich motifs, such as ultrafast trills, despite their susceptibility to acoustic degradation. Ultimately, the global geography of birdsong reflects a spatially varying equilibrium between physical environmental constraints and the biological drive for complex communication.

Acoustic communication is a cornerstone of animal behavior, having evolved into a spectacular diversity of sound signals that mediate survival and reproduction (*1*–*3*). This diversity is iconic in birds, whose songs are shaped by a complex interplay of evolutionary trade-offs and mediate social interactions among individuals (*4*–*7*). Songs must travel effectively through the environment (*8*–*10*), encode specific information for mates or rivals (*2*–*4*, *11*), and remain within the singer's physical capabilities (*12*, *13*), all within the constraints imposed by evolutionary history (*14*). Although the conjunction of these factors has sculpted a global mosaic of bird acoustics (*15*, *16*), a unifying framework explaining its complexity and worldwide distribution is missing. Specifically, it remains unknown how much the global bird soundscape is a historical consequence of phylogenetic radiation relative to shaping by the physics of the biosphere. In this work, we explore song diversity in passerine birds, testing whether their songs converge on acoustic motifs aligned with environmental gradients.

With nearly 6500 species, passerines represent more than 60% of all birds and have radiated over the past 50 million years, colonizing nearly all terrestrial biomes (*17*–*19*). Their global pervasiveness and acoustic richness make them an ideal system for uncovering biogeographic patterns of communication. However, previous large-scale analyses have been hampered by an oversimplified quantification of acoustic complexity, often reducing songs to a few metrics, such as average pitch and frequency range (*15*, *16*, *20*, *21*). These approaches do not fully capture the rich informational content embedded in a song's intricate spectro-temporal architecture—the very patterns of frequency and amplitude modulations that carry its meaning (*22*, *23*).

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In this work, we explicitly treat bird songs as vectors of information to map the global diversity of acoustic communication and uncover its environmental drivers.

We use a holistic framework to understand the composition and worldwide structuring of bird songs. Our study thus integrates the quantification of the relative influence of phylogeny, morphological traits, sexual selection, and social traits on song structure. This is combined with mapping of global song diversity, analysis of associations with environmental gradients (vegetation density, climate, topography, and human footprint), and modeling of song transmission through the environment. We show that the seemingly tremendous acoustic diversity of bird songs can be partitioned into a limited set of elementary "acoustic motifs" that birds combine to construct their songs. We show that the geographic distribution of these motifs follows environmental gradients that structure the composition of regional passerine species assemblages, thereby connecting songbird biogeography and acoustic communication. By providing the geographic map of passerine songs, this work establishes an explicit link between the evolutionary mechanisms shaping communication and the macroecological patterns of global bioacoustic diversity.

Eight acoustic motifs structure the global passerine song diversity

We analyzed the complete spectro-temporal architecture of 116,792 song bouts from 3160 passerine species, encompassing more than half of all extant species and more than 80% of passerine families (fig. S1; see materials and methods for details on automatic song segmentation). We characterized each song bout by its modulation power spectrum, a sound representation that quantifies the modulations of sound energy in time (e.g., trill rates), frequency (e.g., tonality), and joint time-frequency (e.g., sweep curvature) (Fig. 1A and materials and methods) (*24*). A modulation power spectrum captures a sound's spectro-temporal texture in a manner that is both biologically salient, because these modulations are key to how birds encode information, and analytically robust for comparing song patterns across thousands of species (materials and methods and figs. S2 and S3).

We explored the variability of these modulation power spectra in a 37-dimensional acoustic space where proximity between song bouts reflects their spectro-temporal similarity (Fig. 1B and figs. S4 and S5). Using unsupervised classification, we partitioned this high-dimensional volume into eight distinct acoustic motifs. Although the acoustic space of bird songs is continuous, song bouts concentrate around eight poles, each captured by a motif (fig. S6). Motif assignment is unambiguous: 75% of the song bouts are classified as belonging exclusively to one of the eight motifs with low ambiguity (maximum posterior probability > 0.9; mean = 0.92, median = 0.99). These motifs form elementary building blocks for passerine songs (Fig. 1C, fig. S7, and materials and methods). They include three types of trills distinguished by their speed, three types of whistles based on their pitch modulation, and two more complex structures: atonal chaotic notes and harmonic stacks (Fig. 1C, fig. S8, and the interactive web application in the supplementary materials). On average, each species displays 4.6 ± 1.8 (SD) different motifs in its song repertoire; 65% of all analyzed songs are composites that combine two to eight motifs (mean = 2.23 ± 1.25 motifs). Notably, most species (65.3%) display their dominant motif in more than 80% of their songs (fig. S9). In the other species (34.7%), the dominant motif is used to a lesser extent, which allows for greater song variability. Given that passerines share a keen ability to resolve complex spectro-temporal modulations, these eight motifs are certainly perceptually distinguishable by all passerine species (*25*).

Acoustic motifs are shaped by life history beyond shared ancestry

We first tested whether the dominant motif of each species reflects its phylogenetic history. We built a circular phylogram reconstructing the

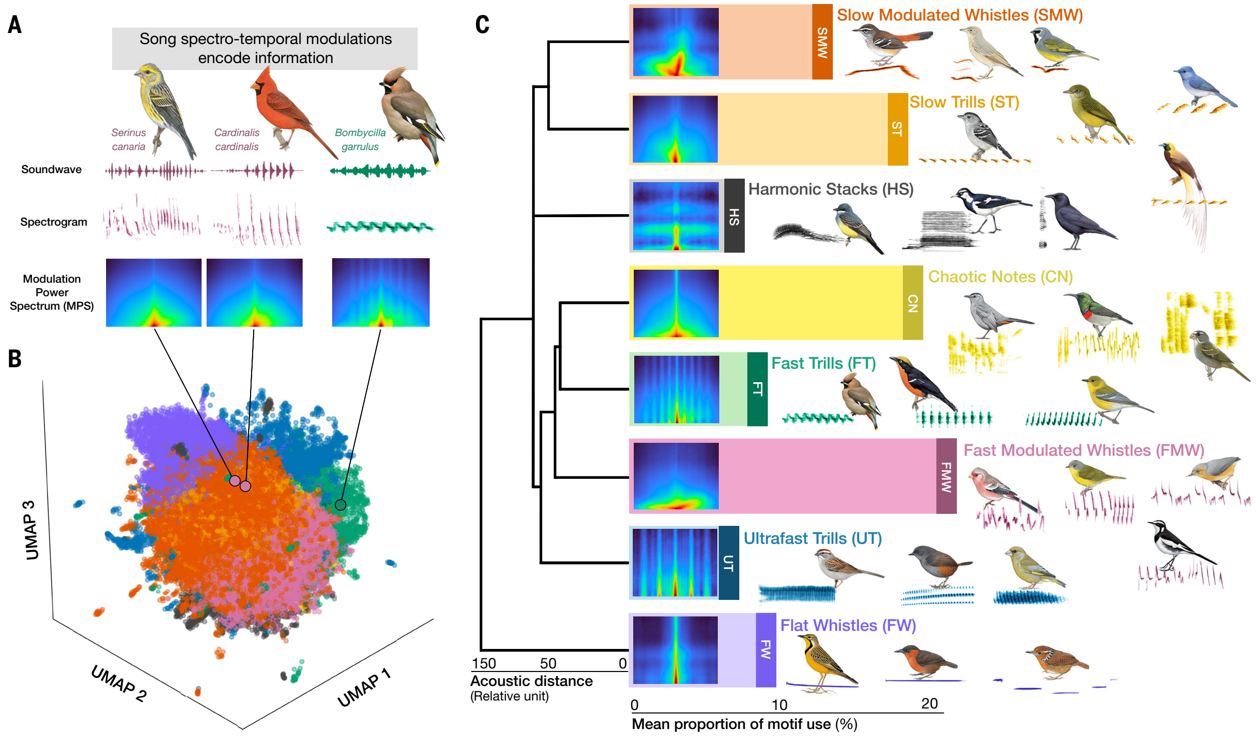

Fig. 1. Eight distinct acoustic motifs structure the global passerine song repertoire. (**A**) Song feature extraction: soundwave, spectrogram, and modulation power spectrum (MPS) (see materials and methods). The MPS quantifies spectro-temporal modulations encoding information such as species identity, individual identity, and singer's quality. Two of the three species shown as examples exhibit songs with similar spectro-temporal patterns (colored in purple) that contrast with the pattern found in the vocalization of the third species (colored in green). (**B**) Global acoustic space derived from 116,792 passerine vocalizations (3160 bird species). Visualization uses a three-dimensional uniform manifold approximation and projection (UMAP) of the 37-dimensional MPS space (materials and methods). Colors represent acoustic motifs. (**C**) The eight acoustic motifs. Dendrogram shows acoustic similarities (Euclidean distances between MPS profiles). Bar plots indicate the proportion of each motif across all passerine songs. Representative MPSs (song bout with probability = 1 for that motif) and spectrograms exemplify each acoustic motif. See also fig. S8 and the interactive web application in the supplementary materials, which visualize and play the global vocal repertoire of passerines. [Bird illustrations: © Lynx Nature Books and courtesy of the Cornell Lab of Ornithology (supplementary materials, list of artists)]

ancestral states of these motifs (Fig. 2A). Our analysis revealed a weak phylogenetic signal (Pagel's $\lambda = 0.17$, Blomberg's $K = 0.08$), consistent with closely related species frequently using different dominant motifs. However, phylogenetic conservatism varied substantially across specific motifs: The simplest tonal elements (flat whistles and slow trills) exhibit the strongest phylogenetic signal (λ = 0.88 and 0.84, respectively), whereas fast trills show the weakest ($\lambda = 0.25$), with intermediate values for other motifs. Consistent with prior knowledge (*4*, *26*–*29*), the phylogenetic signal is weaker in oscine passerines, whose songs are culturally transmitted (*3*–*5*), compared with suboscine passerines, where vocal production is genetically driven (*3*–*5*) (dominant acoustic motif split by oscines versus nonoscines: $\lambda_{\text{oscines}} = 0.09$, $\lambda_{\text{nonoscines}} = 0.25$, Wilcoxon test $P < 0.001$).

Phylogeny accounted for 82% of the explained variation in motif use, leaving 18% for biological traits (multivariate variance partitioning; see materials and methods). Within this trait-based component, social organization accounted for 41% of the variance followed by morphology (34%) and mating system (25%). Given this trait signal, we used a Dirichlet regression to quantify each trait's effect on motif composition (materials and methods and fig. S10).

Social organization, reflecting the propensity of individuals to form long-term associations and produce long-range vocal signals (*30*), drives a marked compositional shift in song motifs (Fig. 2B, top panel, and fig. S11). Species engaging in long-range communication produce a higher proportion of flat whistles and slow trills, reallocating ~5% [4.9%, 95% credible interval (CI): 3.4 to 6.6%] of their repertoire to these motifs compared with species communicating at shorter ranges. By contrast, species signaling at close range favor faster and more complex motifs, such as harmonic stacks and ultrafast trills.

Biophysical constraints imposed by body size and beak shape drive a comparable shift (4.8%, 95% CI: 3.3 to 6.5%) (Fig. 2B, middle panel, and fig. S12). Consistent with prior research in specific avian groups (*31*, *32*), simpler sounds (flat whistles and slow trills) are more represented in larger birds. Conversely, fast modulated whistles, a challenging motif to produce, are predominantly found in smaller species, which may take advantage of their neuromuscular agility to generate complex signals (*13*).

Finally, the mating system drives a smaller but consistent shift in motif composition (3.6%, 95% CI: 2.4 to 4.9%) (Fig. 2B, bottom, and fig. S13). Complex sounds, such as fast modulated whistles, occur more often in polygynous species, a result consistent with the evolution of physiologically costly signals as honest indicators of emitter quality when sexual selection is intense (*2*, *6*, *33*). Conversely, simpler and less demanding motifs, such as flat whistles and slow trills, are more common in monogamous species subject to weaker sexual selection pressures.

The global map of bird songs echoes biogeographic history and biome patterns

We investigated whether the diversity of acoustic motifs used by birds living in the same area varies across the planet by quantifying this diversity within geographical assemblages of bird species found in

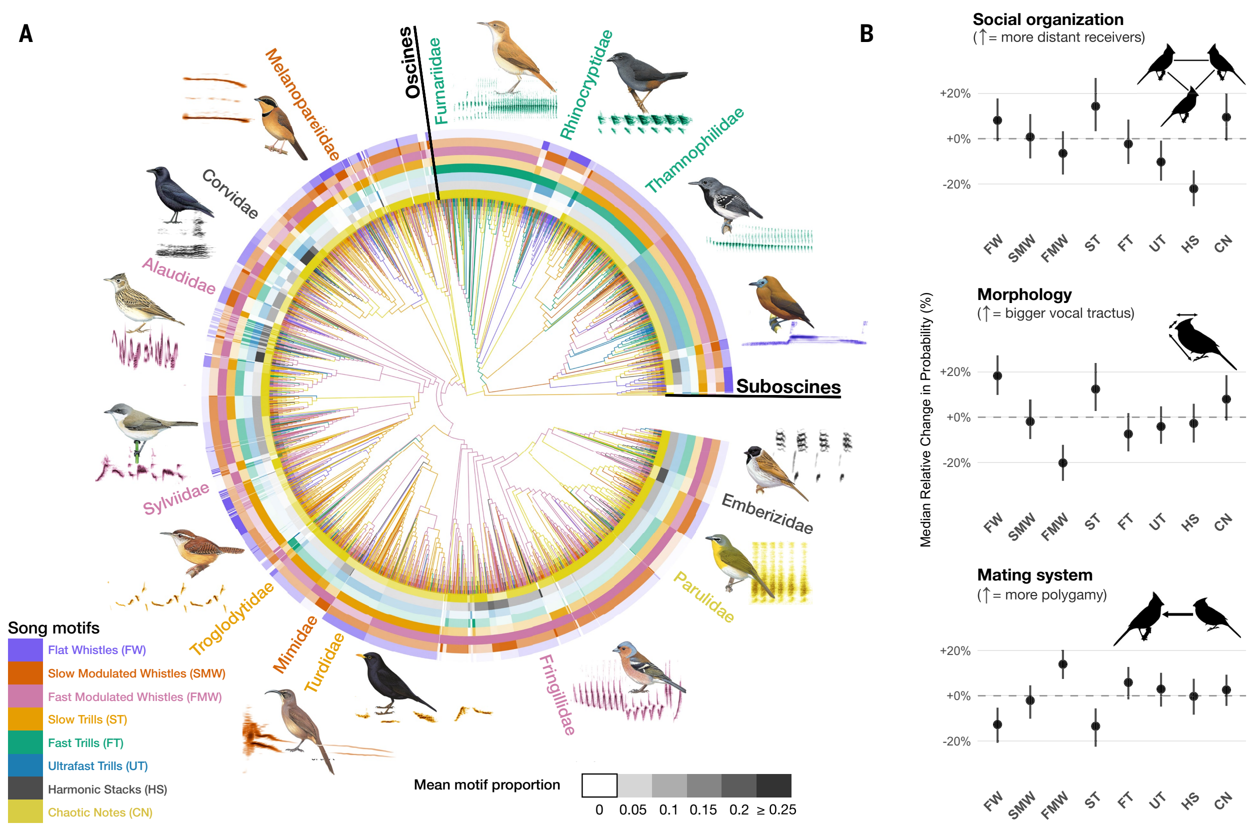

Fig. 2. Phylogenetic distribution and trait correlates of acoustic motifs in passerine birds. (**A**) Passerine phylogeny showing dominant acoustic motifs (ancestral states reconstructed) along branches. Outer heatmap indicates family-aggregated mean of species' posterior probabilities for a given motif. Representative families, species, and spectrograms are shown. [Bird illustrations: © Lynx Nature Books and courtesy of the Cornell Lab of Ornithology (supplementary materials, list of artists)] (**B**) Species traits predict the motifs composing bird songs. Dots show the relative change in each motif's predicted share of the repertoire when comparing species at the 10th versus 90th percentile of each trait. For example, +10% means that a motif becomes 10% more prevalent in the repertoire of high-trait versus low-trait species. FW, flat whistles; SMW, slow modulated whistles; FMW, fast modulated whistles; ST, slow trills; FT, fast trills; UT, ultrafast trills; HS, harmonic stacks; CN, chaotic notes. Error bars show 95% CIs (Dirichlet model with trait gradients as predictors; see materials and methods for details).

$1° \times 1°$ grid cells (Fig. 3A). We found that co-occurring species exhibit more similar acoustic motifs than expected by chance given local species richness [mean functional dispersion measured as a standardized effect size (SES) = −2.1 ± 1.7; Fig. 3A and fig. S14]. This acoustic "niche packing" (*34*) implies that underlying ecological or evolutionary processes drive acoustic convergence within assemblages. Notably, we found no tendency for birds to use a greater number of distinct song motifs than expected by chance in species-rich areas where competition for making oneself heard in acoustic space may be strongest. This lack of trait divergence between co-occurring species suggests that interspecies acoustic competition is insufficient to shape motif usage at broad scales, though it may still occur at finer spatial or temporal resolutions (*11*).

To further investigate how the eight song motifs are geographically structured, we mapped motif usage in every grid cell. More specifically, we counted the number of bird species displaying a given motif in more than 5% of their songs (fig. S15). This observed species count was compared with a null expectation model that randomly drew, for a grid cell, the same number of species from the pool of species occurring in the same biogeographic realm (fig. S16A). Observed counts often departed from the null expectation in geographical regions corresponding to Earth's major biomes (Fig. 3, B to I, and fig. S16B). Notably, song motifs characterized by constant or slowly varying sounds, such as flat whistles and slow trills, are disproportionately common in tropical rainforests. These overrepresented song motifs reach their highest relative prevalence in the Amazonian, Congo Basin, and Southeast Asian rainforests. Biome identity alone explains 47% of intercell variance in flat whistle prevalence and 43% in slow trill prevalence (table S1). Conversely, fast modulated whistles are underrepresented in tropical rainforests. Instead, they occur broadly across the Palearctic, largely independent of biome boundaries within this realm. The global map of the most overrepresented motif for each location (fig. S16C) confirms these observations. Only slow modulated whistles and harmonic stacks are largely independent of biome identity [coefficient of determination (R^2) = 0.18 and 0.19, respectively; table S1]. These patterns reveal that some motifs track broad habitat types across continents, which suggests that ecological constraints may shape bird acoustic signals at a planetary scale.

Global variation in climate and vegetation shapes the efficiency of song transmission

One of the most important ecological constraints that may explain the geographical distribution of acoustic motifs could be the effect of the environment on sound information degradation. The physical environment degrades sound waves during propagation, imposing strong selection on acoustic signals to maintain information transfer (*8*, *9*). However, propagation-induced constraints depend strongly on the properties of the transmission channel—e.g., presence and density of vegetation (*10*). We thus tested whether these environmental pressures contribute to shaping the global arrangement of acoustic motifs. We

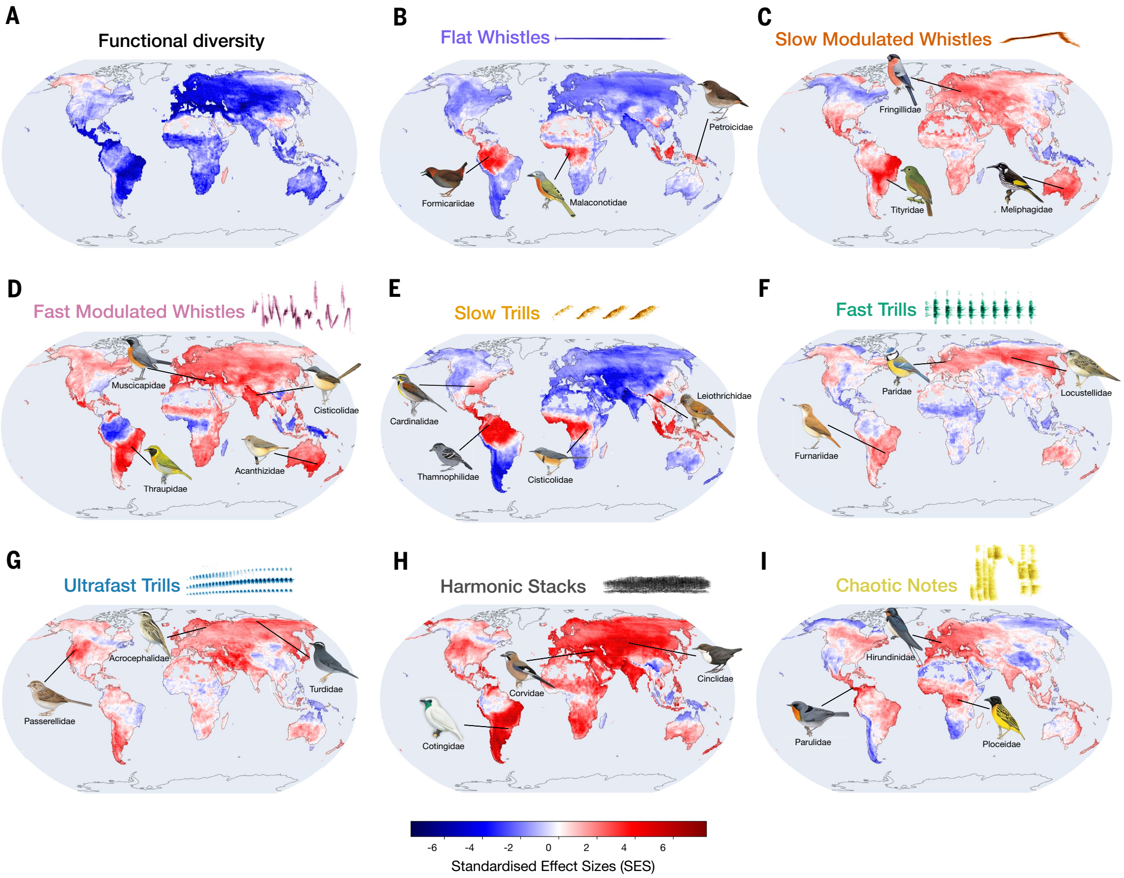

Fig. 3. The geographic structure of songbird acoustic motifs is partly consistent with terrestrial biomes. (**A**) Diversity of acoustic motifs within geographical assemblages of bird species [1° × 1° grid cell; diversity quantified by SESs of functional dispersion (FDis)]. Dark blue indicates less motif diversity in acoustic assemblage than expected by chance given local species richness, and red indicates overdispersed assemblages. (**B** to **I**) Deviation maps of motif-specific species richness per 1° × 1° grid cell. Values are SESs that quantify the deviation in motif species richness from a null expectation based on random sampling. A positive SES indicates an overrepresented motif given local species richness, whereas a negative SES indicates an underrepresented motif given local species richness. Representative bird species are shown in areas where a given motif largely exceeds its expected species richness (top 5% SES). [Bird illustrations: © Lynx Nature Books and courtesy of the Cornell Lab of Ornithology (supplementary materials, list of artists)]

developed a physics-based transmission efficiency index (TEI) to assess the ability of acoustic motifs to convey species identity information along environmental gradients affecting sound propagation (materials and methods). This index quantifies the persistence of a signal's informational content during propagation by modeling information degradation caused by key physical processes in a forest environment (fig. S17). These processes include attenuation (overall sound intensity loss), reverberation (echoes filling the internote silences and blurring the sound structure), frequency-dependent filtering (greater loss of signal energy for higher-pitched sounds), and ambient noise (background sounds that mask the signal) (figs. S17 and S18) (*3*, *4*, *10*). As expected, given their simpler acoustic structure (*4*), flat whistles and slow trills exhibit the highest TEI values, which highlights their resilience to propagation-induced information degradation (Fig. 4A and fig. S19). By contrast, spectro-temporally complex signals, such as harmonic stacks and chaotic notes, are least effective in transferring information over long ranges. To refine our understanding of environmental constraints on sound propagation, we further simulated propagation in an open habitat, where reverberation due to vegetation is absent but where atmospheric turbulence contributes to signal masking (*35*) (materials and methods). Under these open-habitat conditions, information transmission is more efficient, regardless of the song motif considered (mean TEI in open habitat = −0.36 versus −0.77 in closed habitat; TEI dispersion reduced by 41%; fig. S20). This result emphasizes that propagation constraints in densely vegetated environments constitute a differential selective pressure among birdsong motifs.

Mapping mean TEI values at a planetary scale reveals a distinct signature of rainforests (Fig. 4B). Grid cells across Amazonia, the Congo Basin, and Southeast Asia are significantly enriched in motifs associated with long-range information transmission and depleted in motifs characterized by a lower transmission index. To disentangle the environmental drivers of this geographic pattern, we correlated the TEI with standardized environmental predictors. The index shows strong positive correlations with vegetation structure [Pearson's correlation coefficient (r) = 0.50; $P < 0.001$], temperature ($r = 0.36$; $P < 0.001$), and relative humidity ($r = 0.31$; $P < 0.001$) (Fig. 4C). Overall, acoustic motifs most suitable for long-distance information transfer are thus used by more bird species than predicted by chance in densely vegetated environments with high humidity and temperature, mostly located in tropical rainforests. Using spatial multiple regressions including all environmental drivers alongside phylogenetic diversity, we found that humidity is the strongest positive driver of the TEI (Fig. 4D and table S2). A 1-SD increase in humidity raises the index by ~+0.55 SD units (+0.28 accounting for phylogenetic diversity). Temperature also shows a positive effect (+0.10 without versus +0.38 with phylogeny). The effect of vegetation is smaller but still positive (+0.07 without

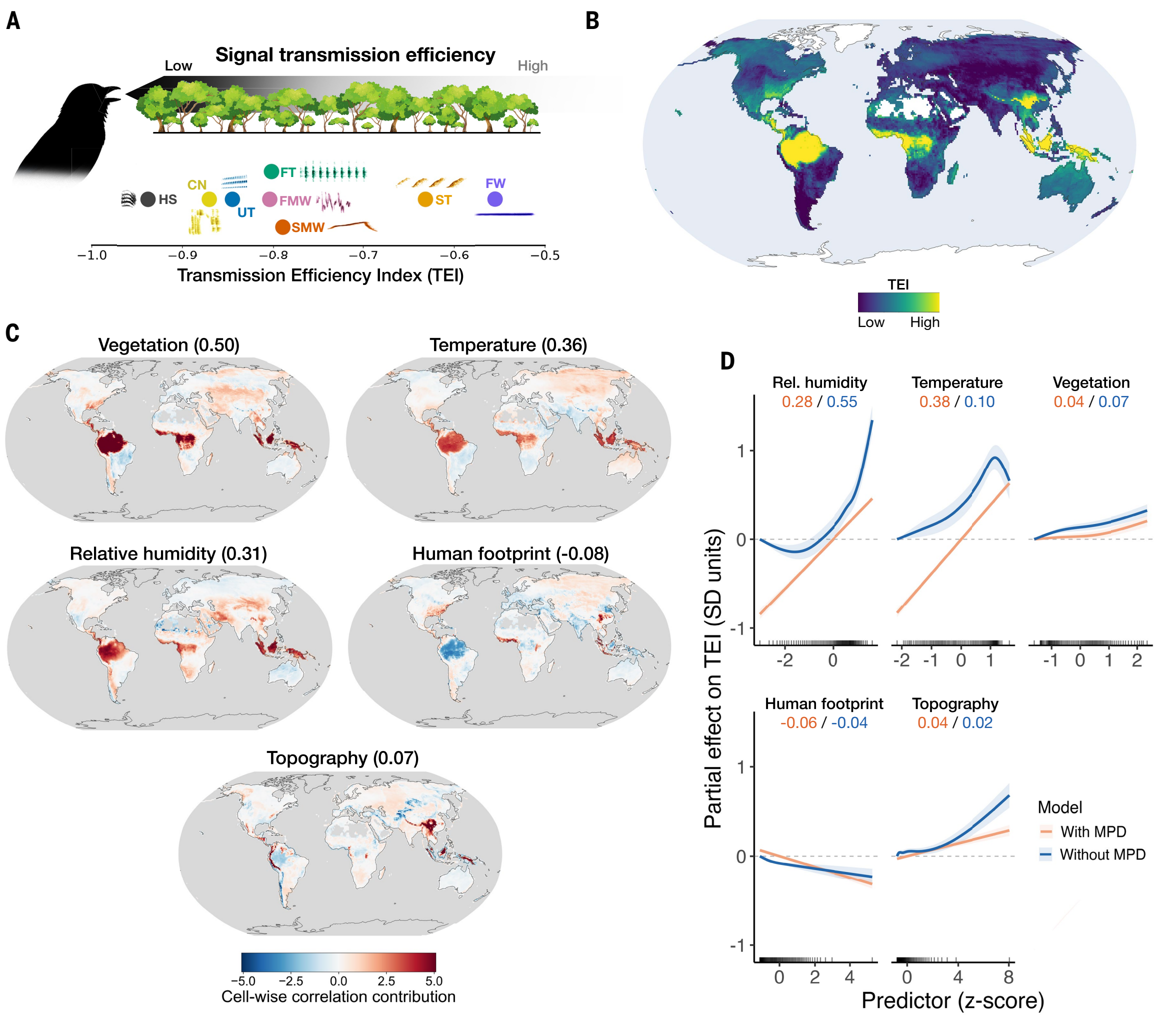

Fig. 4. Climate and vegetation modulate global acoustic transmission efficiency. (**A**) The TEI quantifies an acoustic motif's robustness to information loss during propagation through the environment. Acoustic motifs, such as flat whistles, preserve species identity over farther distances compared with harmonic stacks (see materials and methods). (**B**) Global distribution of TEI. Each 1° cell's value is the mean TEI in the species assemblage. White indicates no data. (**C**) Maps showing the covariation score *c* between TEI and each predictor *x* per 1° cell *i*: $c_i = z(\mathrm{TEI}_i)z(x_i)$. *c* values are positive where TEI and *x* are both above or below their means and negative where they move in opposite directions. The Pearson correlation *r* computed across all cells is shown in parentheses above each map. (**D**) The TEI depends on environmental characteristics. Each panel shows the effect of one *z*-scored predictor on TEI (in TEI SD units) with other predictors at their mean (*z* score = 0). Solid lines represent two models: blue excludes mean phylogenetic distance (MPD), and orange includes it. Shaded bands are 95% uncertainty intervals. Black tick marks along the bottom indicate observation locations. Values above each panel are standardized average marginal effects (average TEI change in SD units per 1-SD increase).

versus +0.04 with phylogeny). These results are consistent with a macroecological expression of the acoustic adaptation hypothesis, which proposes that the acoustic properties of animal vocalizations are adapted to habitat constraints facilitating information transfer between individuals (*8*, *36*–*38*). However, although our results support the acoustic adaptation hypothesis in tropical rainforests, where species preferentially use song motifs resilient to signal degradation, they reveal a contrasting regime in temperate latitudes. In these regions, more complex and degradation-prone motifs are favored. We propose that this shift is driven by shorter interindividual communication distances (*39*) and more intense sexual selection (*40*), which together outweigh the costs of propagation constraints.

Conversely, other variables have smaller effects, including topography (+0.04 with versus +0.02 without phylogeny) and phylogenetic diversity (+0.10). Notably, the human footprint [an indicator of the imprint of all human activities on land (*41*)] is associated with lower transmission efficiency (−0.06 with versus −0.04 without phylogeny), a finding consistent with previous studies reporting negative associations between humans and ecological functions at macroecological scales (*42*). This pattern is particularly prominent in regions with a long history of human influence, such as Europe and India (Fig. 4C) (*43*).

Conclusions

Our research unveils a simplicity underlying the spectacular diversity of bird songs: Across thousands of lineages and biomes, passerine songs can be partitioned into a limited set of eight elementary acoustic motifs. We demonstrate that the global distribution of these motifs is structured by three main forces: biological traits that shape species- specific vocalizations; biogeographic history, which influences song diversity in local communities; and the physics of sound propagation, which favors specific motifs in specific environments. Furthermore, distinct motif usage across species with shared phylogeny indicates that the capacity to produce these extant acoustic motifs emerged early in the passerine radiation. Natural selection, potentially driven by the physics of sound transmission, could have fostered rapid and convergent evolution.

Our framework provides a robust foundation for upscaling to a global level the evolutionary hypotheses on signal transmission that were previously formulated at local scales. These results align with functional biogeography's central tenet—that functional traits often converge on a few elementary combinations (*44*). The planetary-scale biogeographic structure of the birdsong fundamental acoustic motifs seems critically shaped by acoustic adaptation to long-distance information transfer (*8*, *36–38*, *45*). However, the adaptive benefits of propagation efficiency are counterbalanced by the demands of information encoding. In agreement with the predictions of the mathematical theory of communication (*46*), complex motifs, such as fast modulated whistles and fast trills, carry more information [e.g., species identity (fig. S21), group and individual identities (*47–50*), population dialects (*25*, *51*), and singer quality (*52*)], but they are highly susceptible to degradation. By contrast, simpler motifs, such as flat whistles and slow trills, convey less information (fig. S21) but are more resilient to propagation (*53*). This trade-off between encoding more information and transmitting it over long distances exhibits strong geographic variation. Climatic stability in tropical rainforests favors large territories and low encounter rates (*39*), requiring resilient, long-range signals. Contrastingly, the pronounced seasonality of temperate environments reduces both territory sizes and breeding periods, intensifying sexual selection (*20*, *40*). With individuals communicating at closer ranges and under strict receiver assessment, selection favors information-dense motifs that are inherently more prone to degradation during propagation. The biogeography of birdsong thus emerges as a spatially varying equilibrium between these two fundamental constraints.

By providing a conceptual framework akin to a "periodic table" of birdsong at a planetary scale (*54*), our work substantiates a general principle: The evolutionary trajectories of animal vocal signals are channeled along a few geographically structured pathways. Expanding this hypothesis beyond bird songs, by testing it across other acoustic signals and taxonomic groups, will provide a solid basis for a theory of animal communication rooted in the interaction between informa-tion coding, rapid signal evolution, biogeography, and functional ecol-ogy (*55–57*).

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SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.aee6239

Materials and Methods; Supplementary Text; Figs. S1 to S23; Tables S1 and S2; References (*58–91*); MDAR Reproducibility Checklist

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The global biogeography of passerine songs

Quentin Bacquelé, Jean-Yves Barnagaud, Cyrille Violle, Frédéric Theunissen, and Nicolas Mathevon

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Editor's summary
Birds sing to attract mates and ward off rivals. The form of these songs can vary from single, repeated notes to complex melodies. Bacquelé *et al*. explored the relationships among song, environment, and latitude from birds across the world (see the Perspective by Briefer). They found that in regions where forests are thick, and thus sound transmission is slower, bird songs are simpler and more effective for transmission across vegetation-impeded landscapes. By contrast, in temperate regions, where sound is less impeded and sexual selection is especially strong due to seasonal limitations, birdsong is much more complex, likely being a better indicator of singer condition. —Sacha Vignieri

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